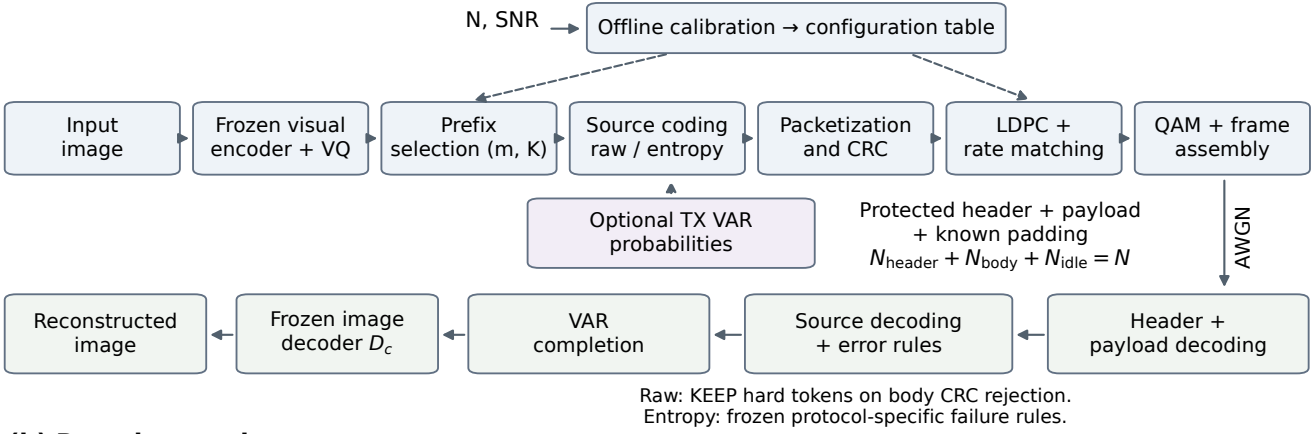
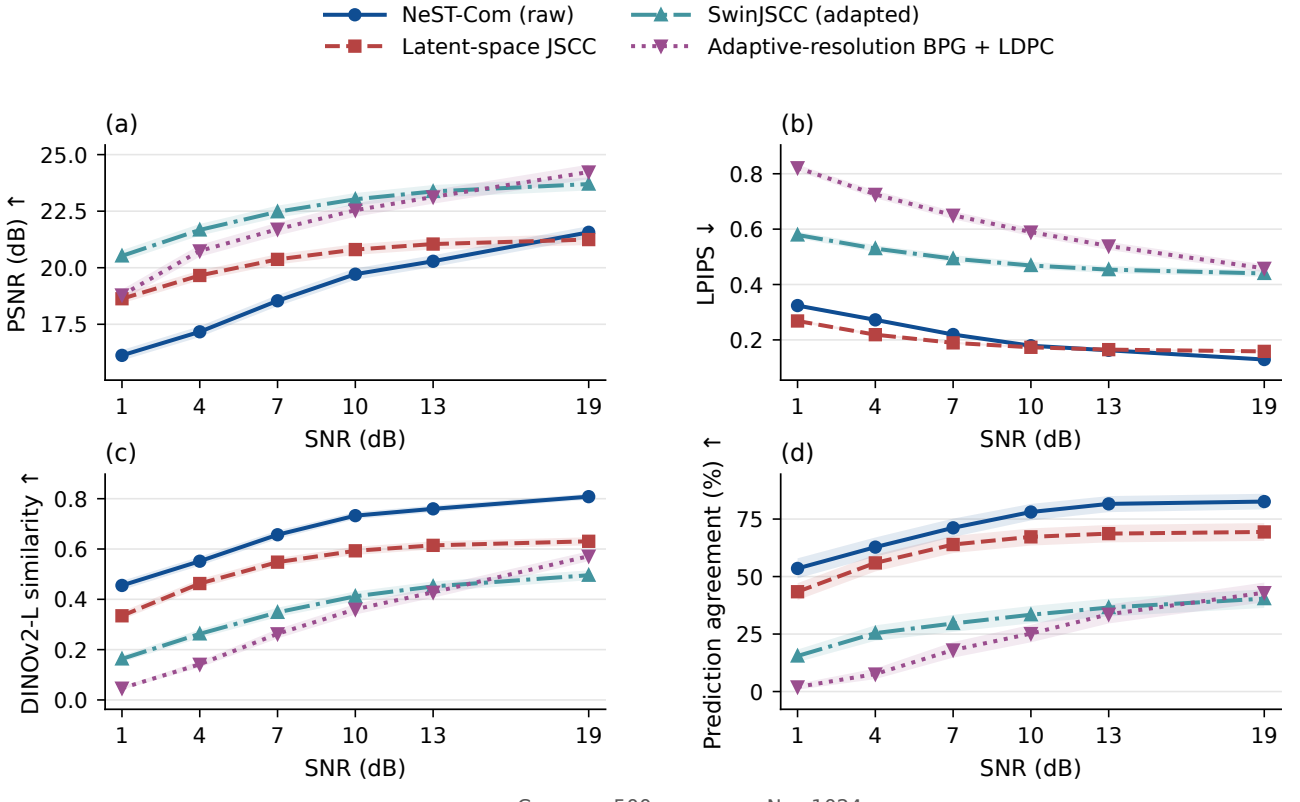
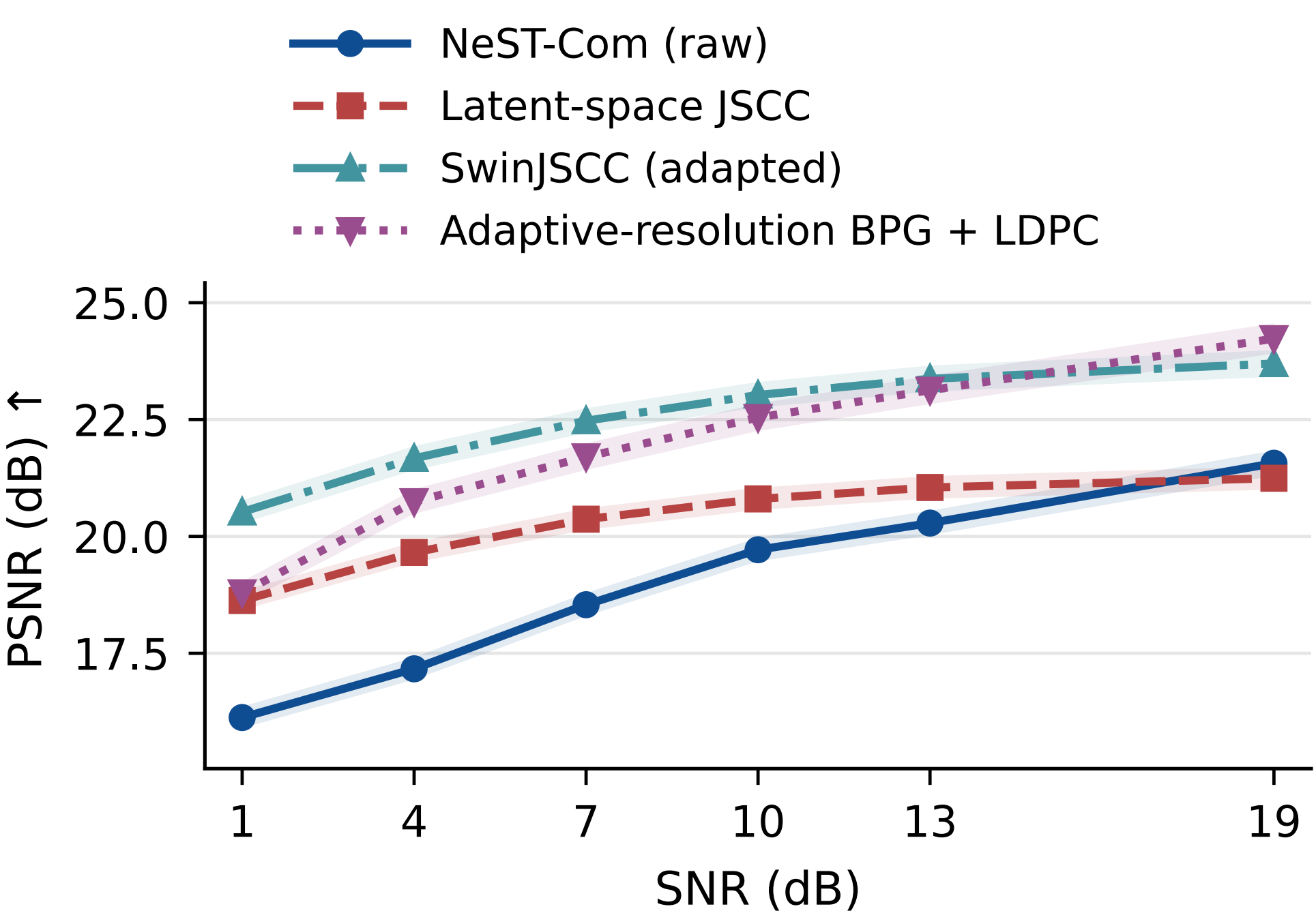
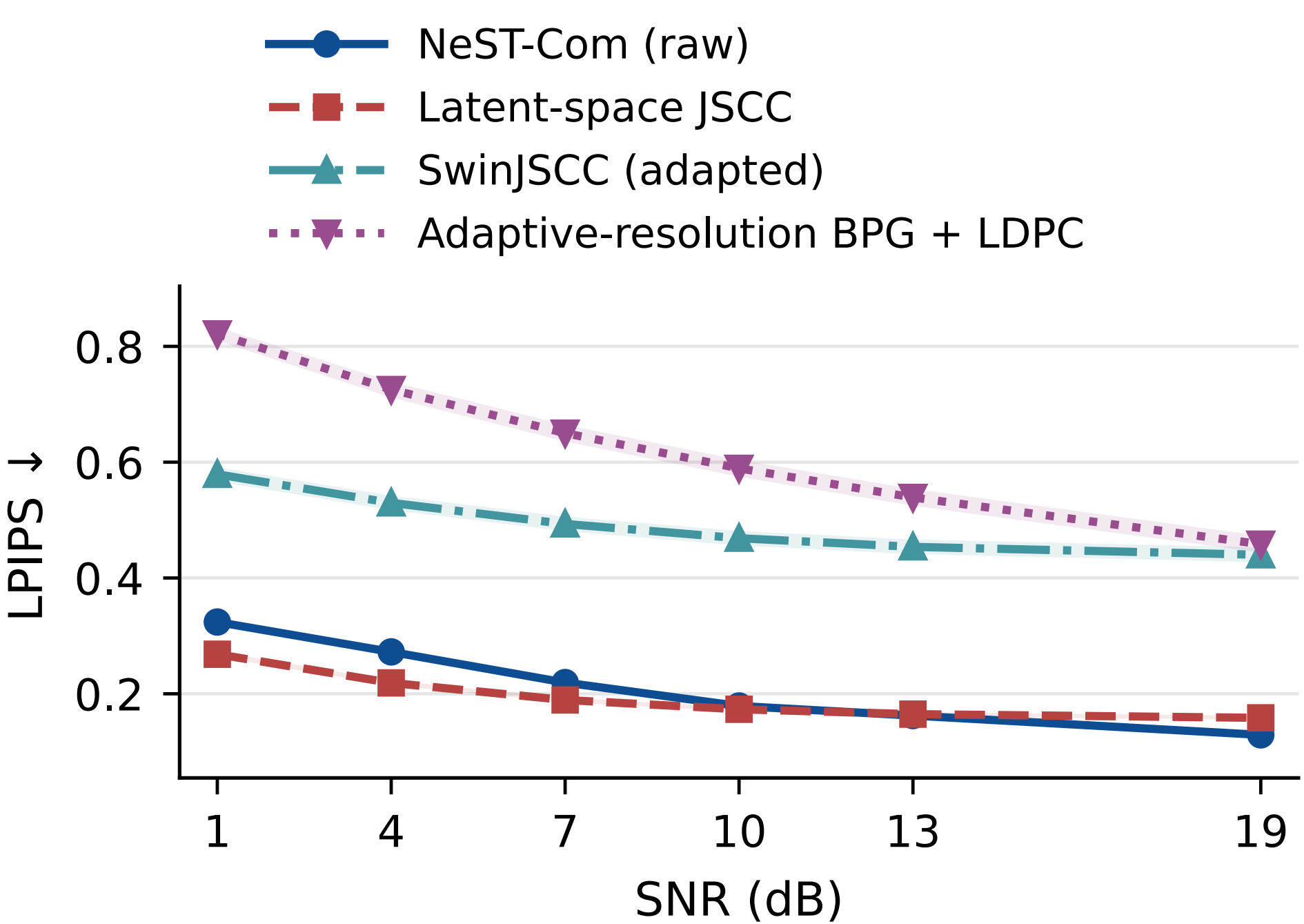

(a) Transmission and reconstruction
N, SNR
Offline calibration → configuration table
Input image
Frozen visual encoder + VQ
Prefix selection (m, K)
Source coding raw / entropy
Packetization and CRC
LDPC + rate matching
QAM + frame assembly
Optional TX VAR probabilities
Protected header + payload + known padding
$N_{header} + N_{body} + N_{idle} = N$
AWGN
Reconstructed image
Frozen image decoder D_c
VAR completion
Source decoding + error rules
Header + payload decoding
Raw: KEEP hard tokens on body CRC rejection.
Entropy: frozen protocol-specific failure rules.
(b) Boundary-scale recovery
Received coarse scale prefix
Predict boundary scale
Overwrite received boundary positions
Update accumulated latent
Predict finer scales
Received tokens condition finer scales; other positions in the same boundary scale are not recomputed.

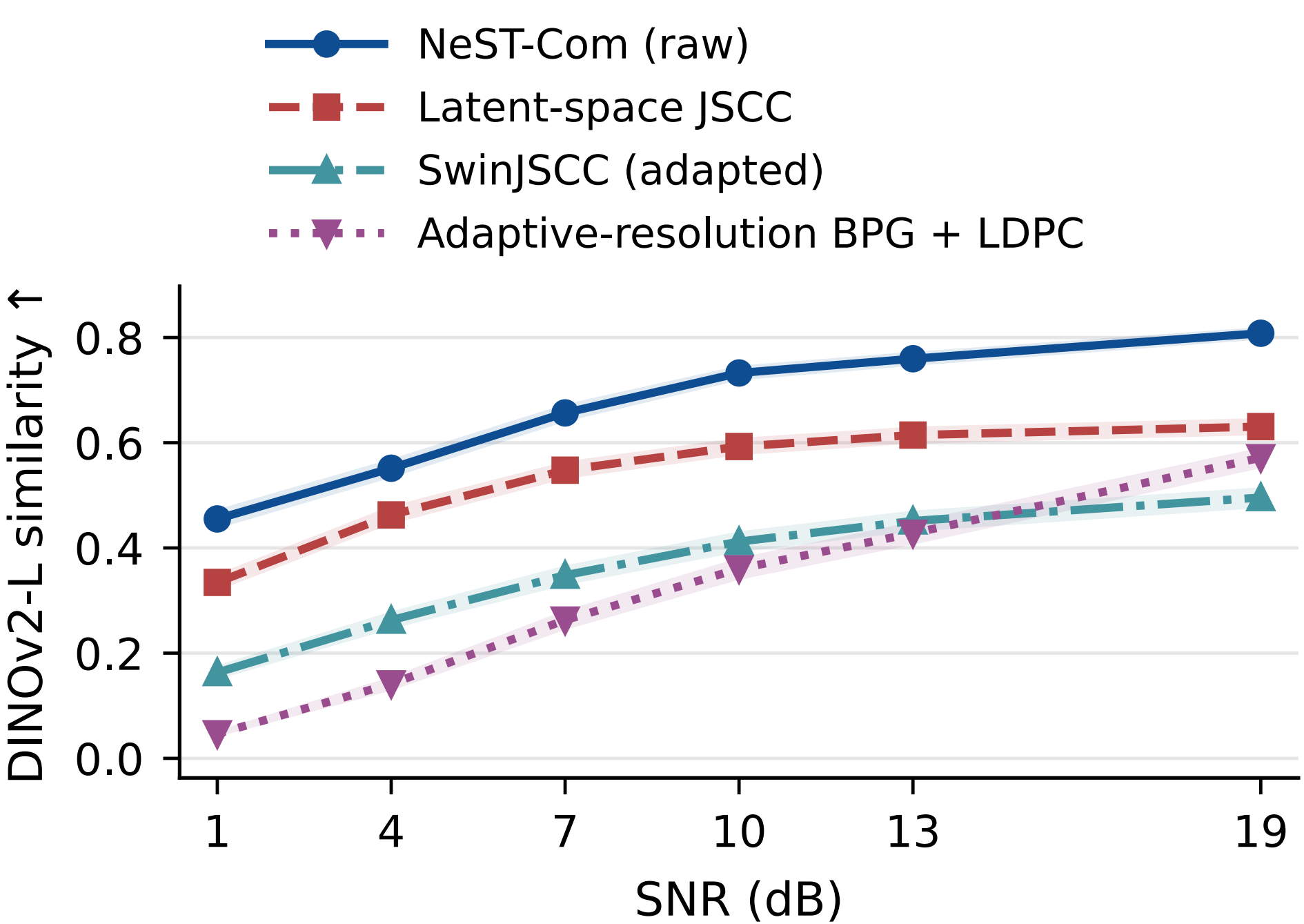

NeST-Com (raw)
Latent-space JSCC
SwinJSCC (adapted)
Adaptive-resolution BPG + LDPC
(a)
PSNR (dB) ↑
25.0
22.5
20.0
17.5
1
4
7
10
13
19
SNR (dB)
(b)
LPIPS ↓
0.8
0.6
0.4
0.2
SNR (dB)
(c)
DINOv2-L similarity ↑
0.8
0.6
0.4
0.2
0.0
SNR (dB)
(d)
Prediction agreement (%) ↑
75
50
25
0
SNR (dB)
Common 500 sources · N = 1024



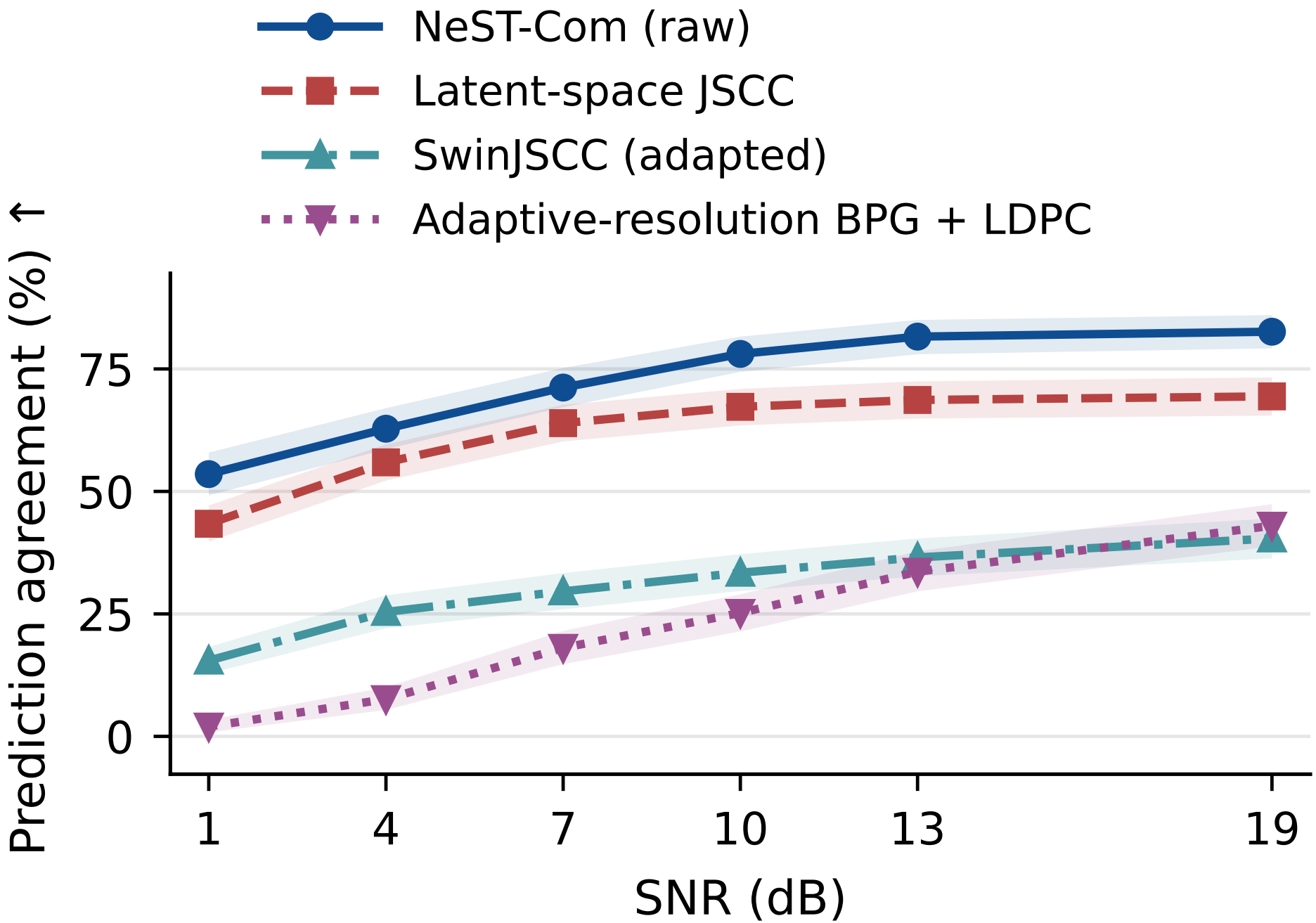
Common 500 sources · N = 1024



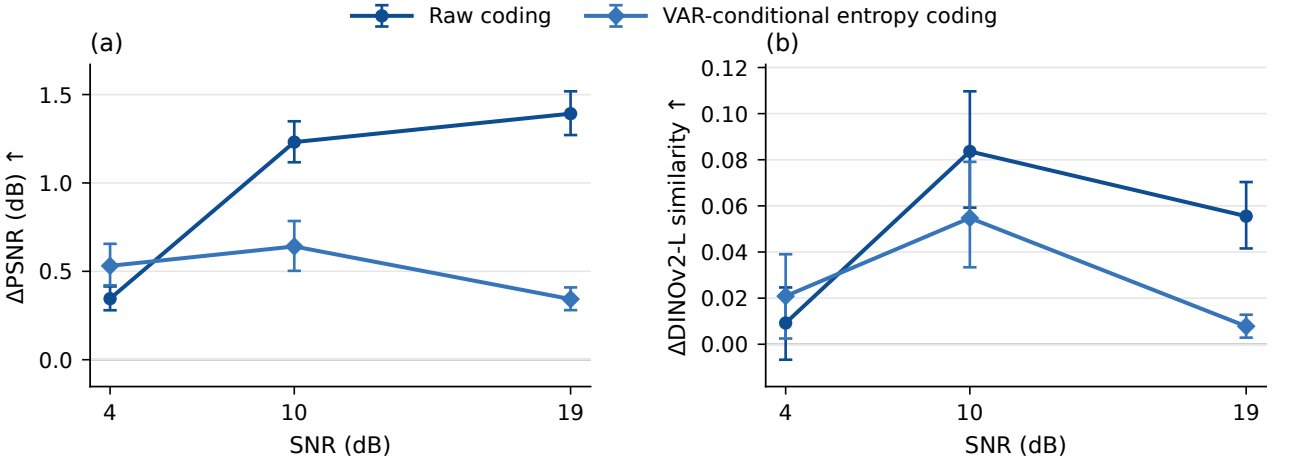
Common 500 sources · N = 1024

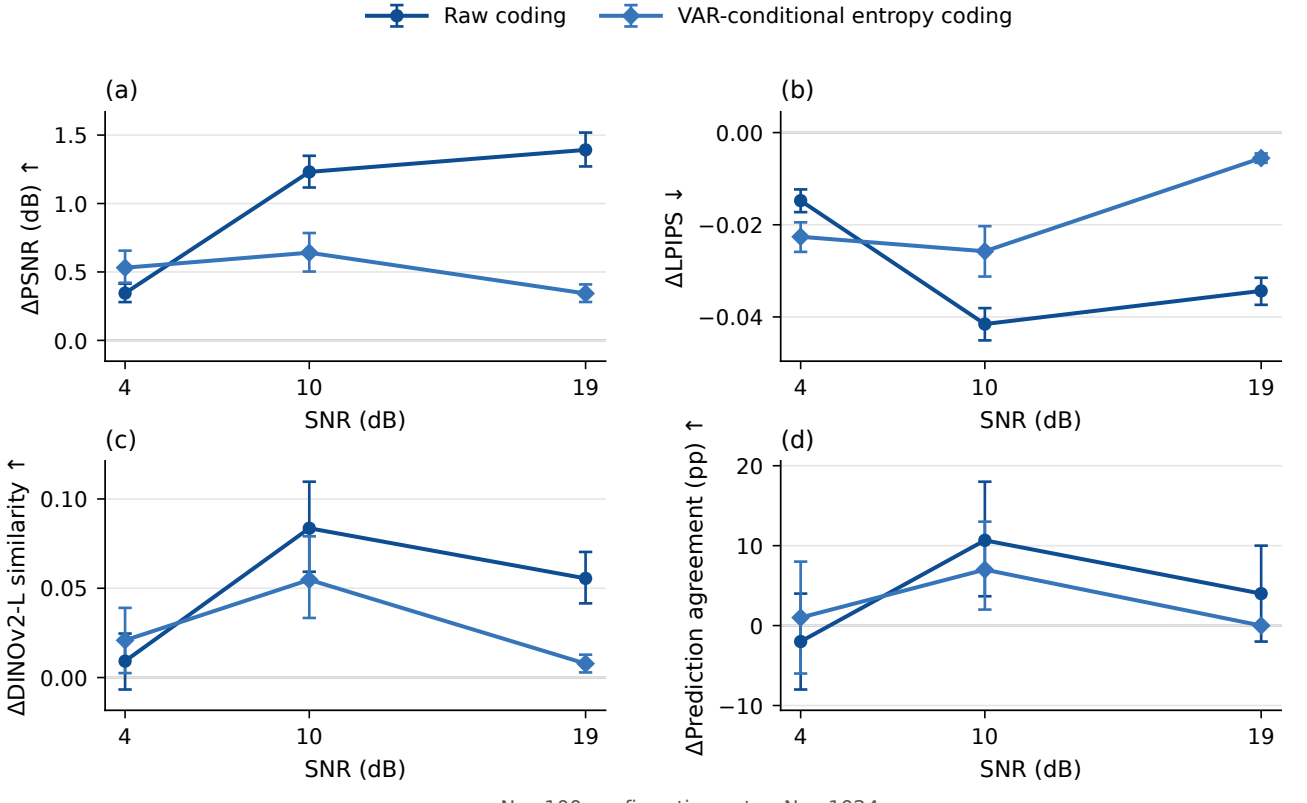


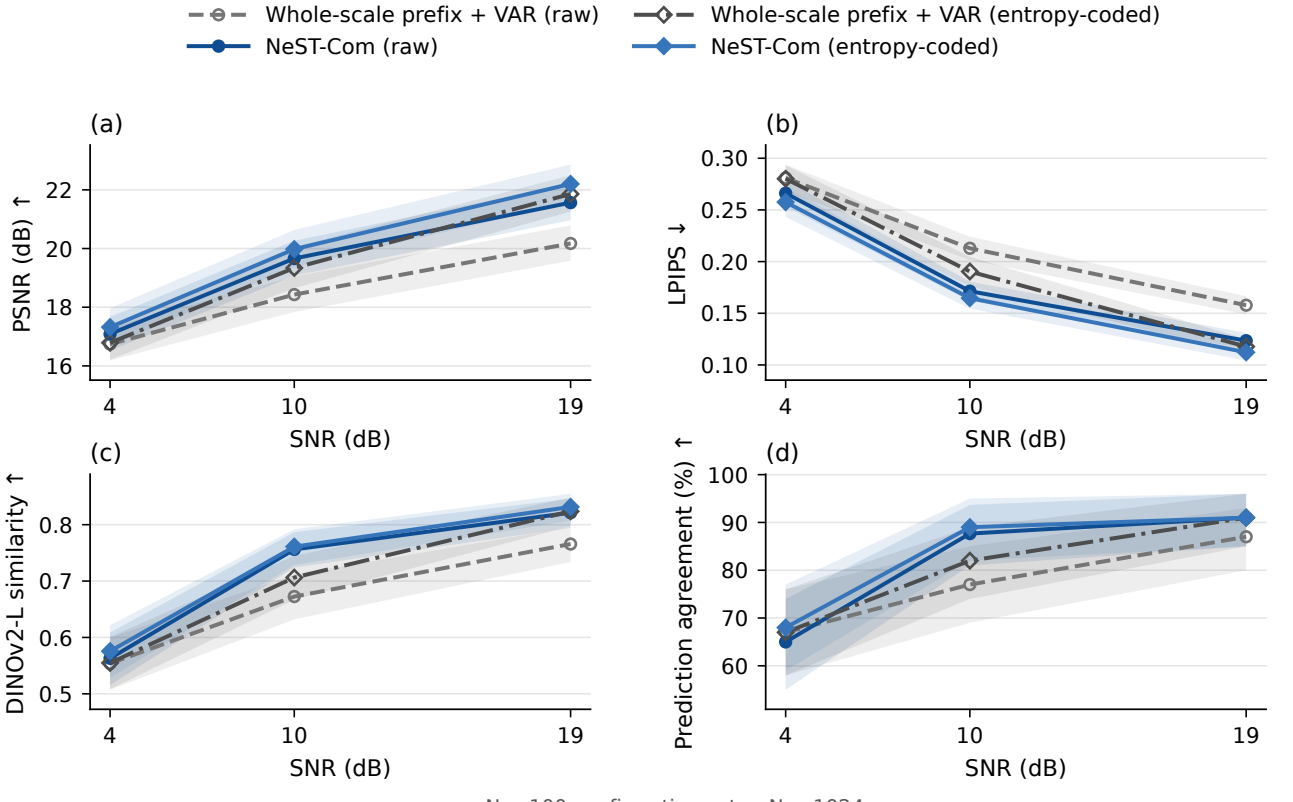
Common 500 sources · N = 1024

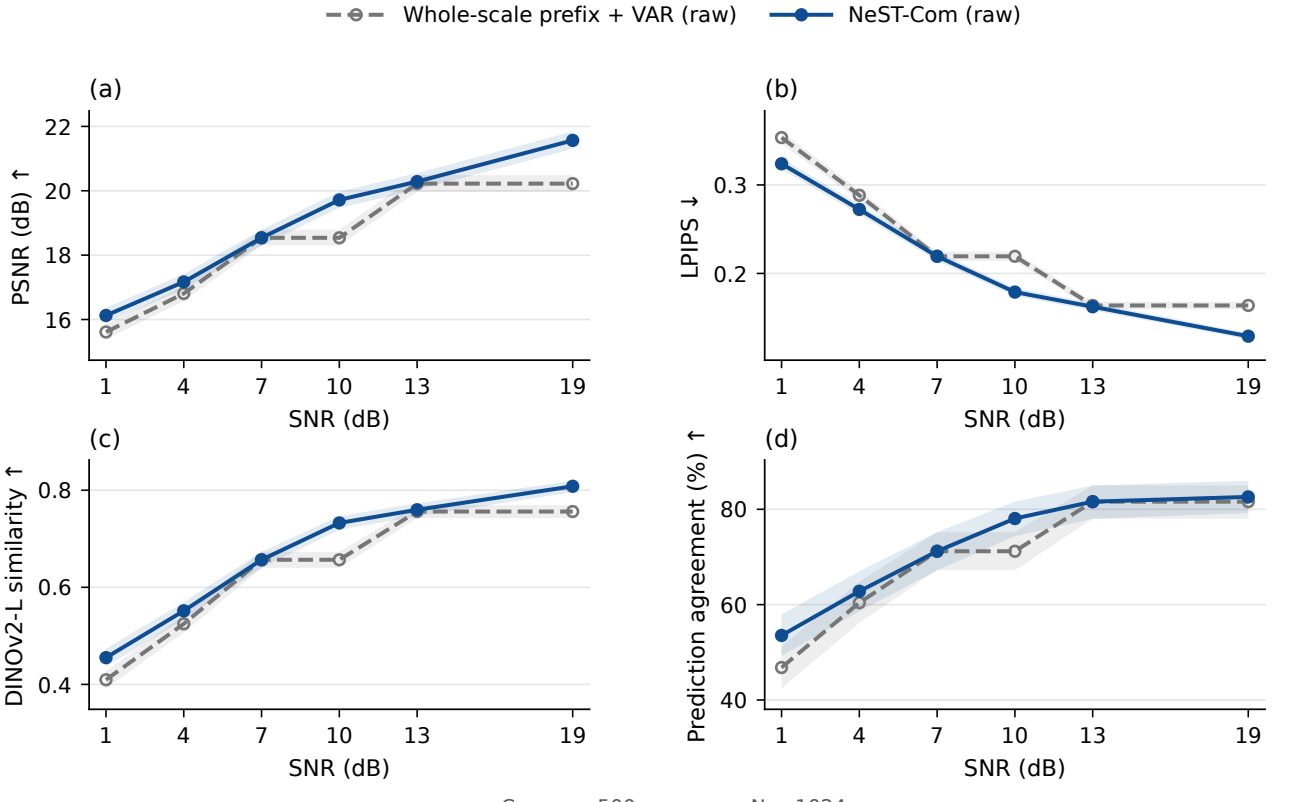


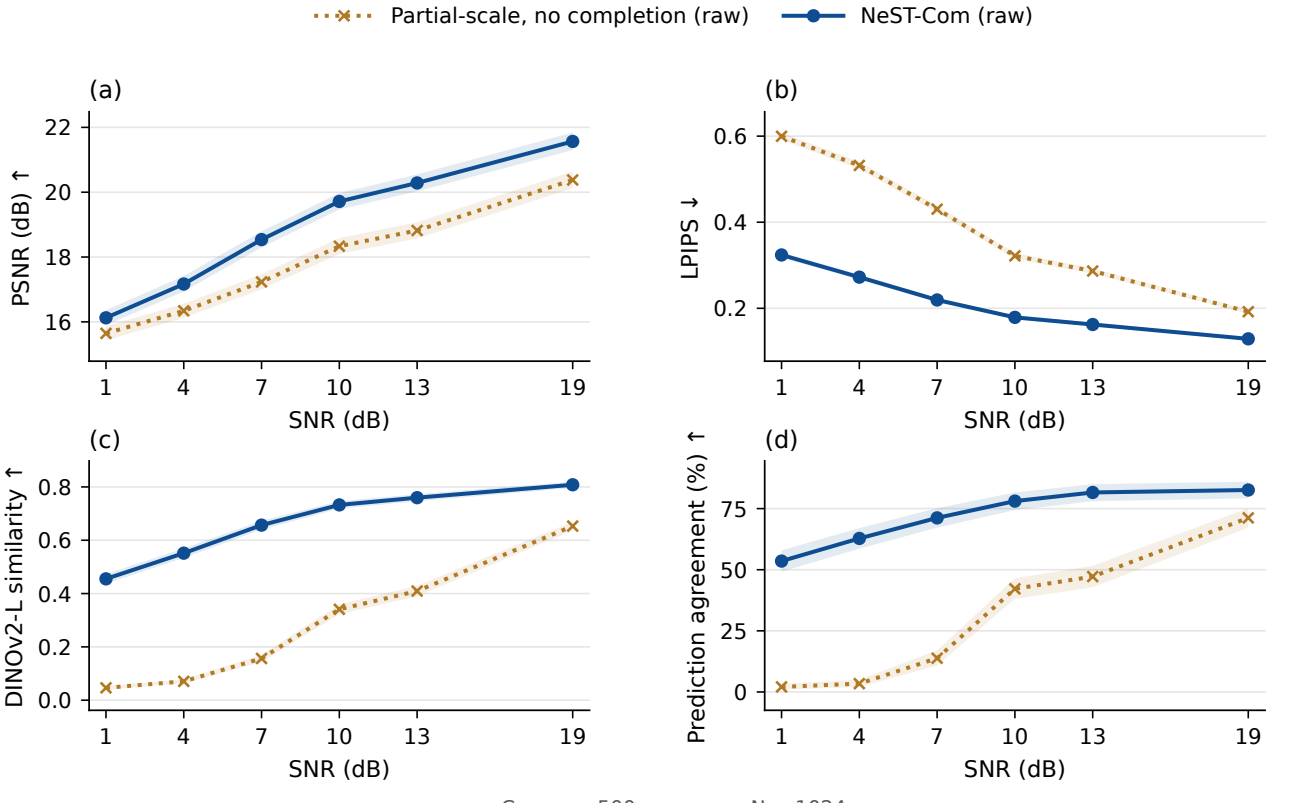
Common 500 sources · N = 1024

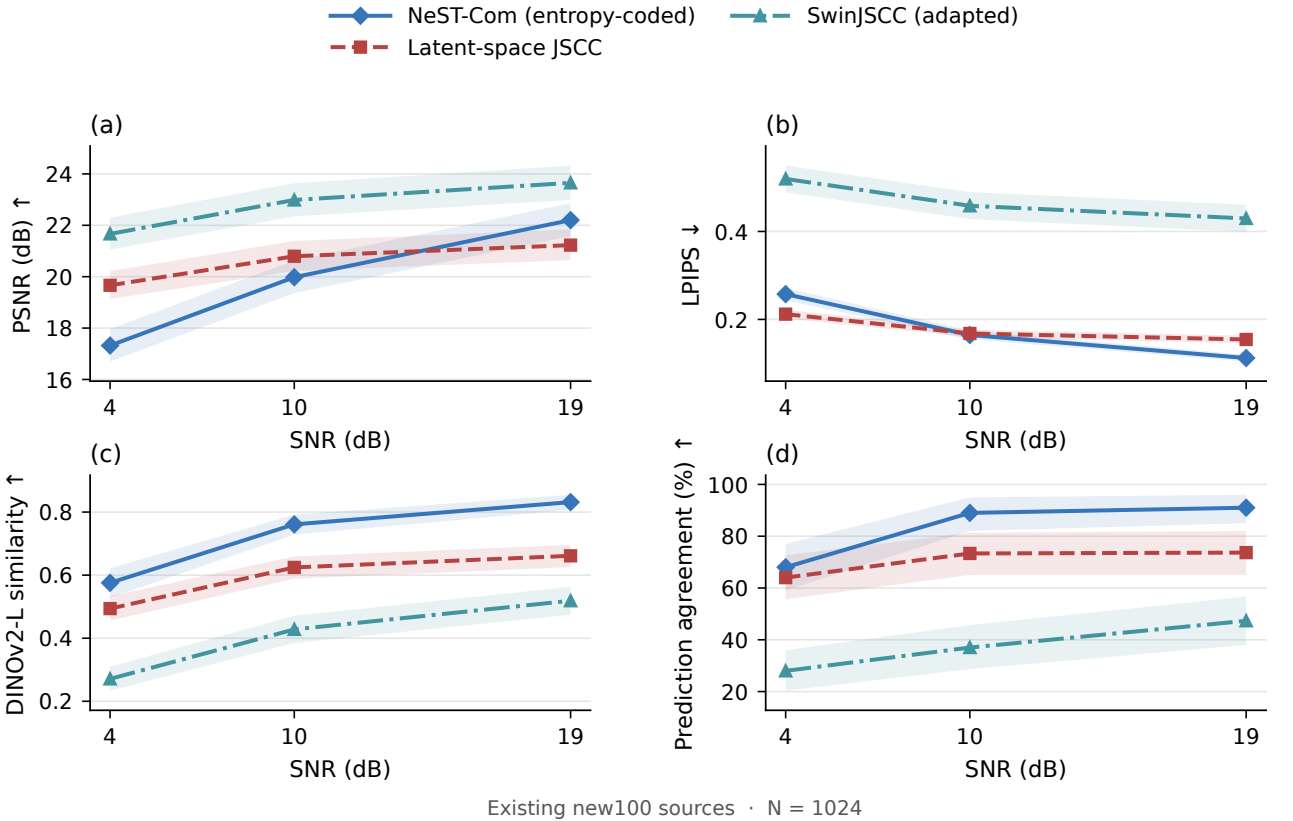

Raw coding
VAR-conditional entropy coding
(a)
ΔPSNR (dB) ↑
1.5
1.0
0.5
0.0
4
10
19
SNR (dB)
(b)
ΔDINOv2-L similarity ↑
0.12
0.10
0.08
0.06
0.04
0.02
0.00
4
10
19
SNR (dB)
New100 confirmation set · N = 1024

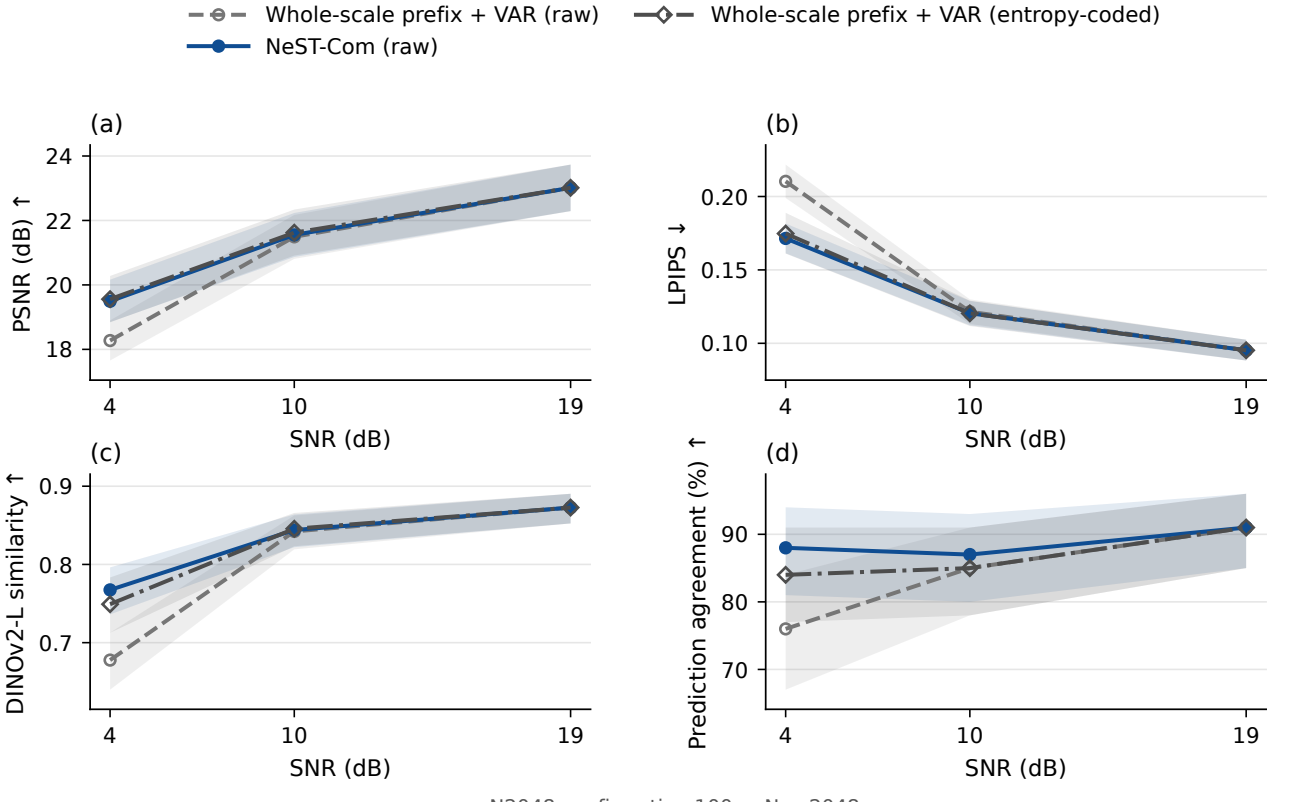

Raw coding
VAR-conditional entropy coding
(a)
ΔPSNR (dB) ↑
1.5
1.0
0.5
0.0
4
10
19
SNR (dB)
(b)
ΔLPIPS ↓
0.00
−0.02
−0.04
4
10
19
SNR (dB)
(c)
ΔDINOv2-L similarity ↑
0.10
0.05
0.00
4
10
19
SNR (dB)
(d)
ΔPrediction agreement (pp) ↑
20
10
0
−10
4
10
19
SNR (dB)
New100 confirmation set · N = 1024

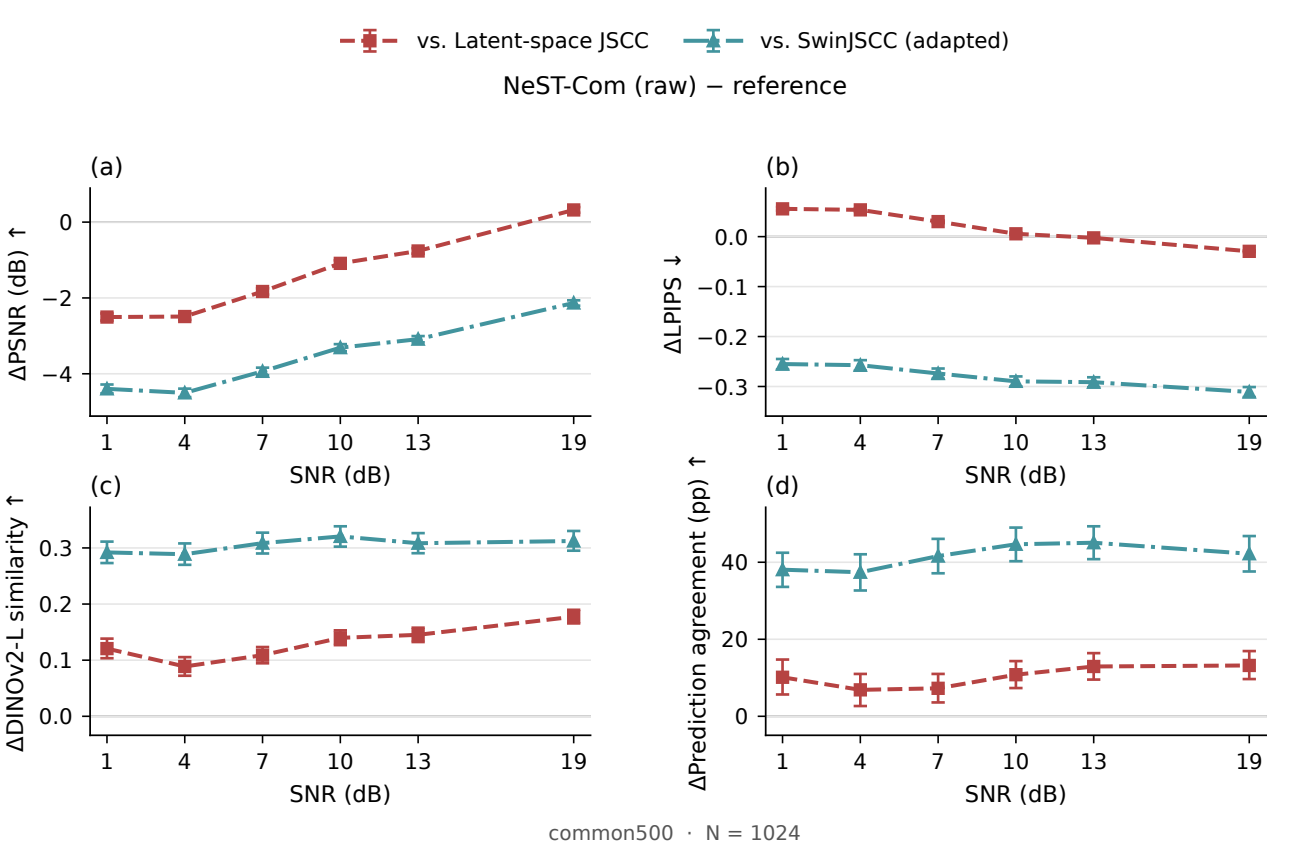

Whole-scale prefix + VAR (raw)
Whole-scale prefix + VAR (entropy-coded)
NeST-Com (raw)
NeST-Com (entropy-coded)
(a)
PSNR (dB) ↑
22
20
18
16
4
10
19
SNR (dB)
(b)
LPIPS ↓
0.30
0.25
0.20
0.15
0.10
4
10
19
SNR (dB)
(c)
DINOv2-L similarity ↑
0.8
0.7
0.6
0.5
4
10
19
SNR (dB)
(d)
Prediction agreement (%) ↑
100
90
80
70
60
4
10
19
SNR (dB)
New100 confirmation set · N = 1024

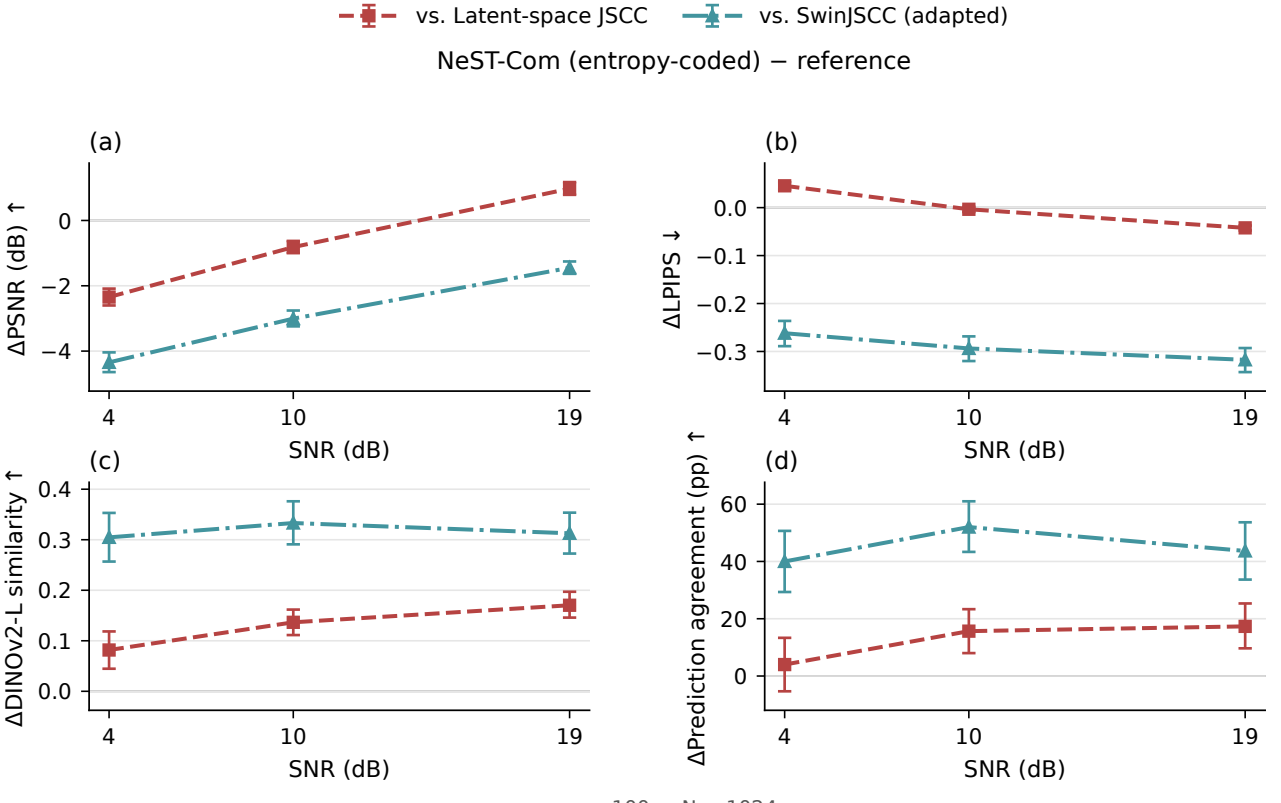

Whole-scale prefix + VAR (raw)
NeST-Com (raw)
(a)
PSNR (dB) ↑
22
20
18
16
1
4
7
10
13
19
SNR (dB)
(b)
LPIPS ↓
0.3
0.2
1
4
7
10
13
19
SNR (dB)
(c)
DINOv2-L similarity ↑
0.8
0.6
0.4
1
4
7
10
13
19
SNR (dB)
(d)
Prediction agreement (%) ↑
80
60
40
1
4
7
10
13
19
SNR (dB)
Common 500 sources · N = 1024

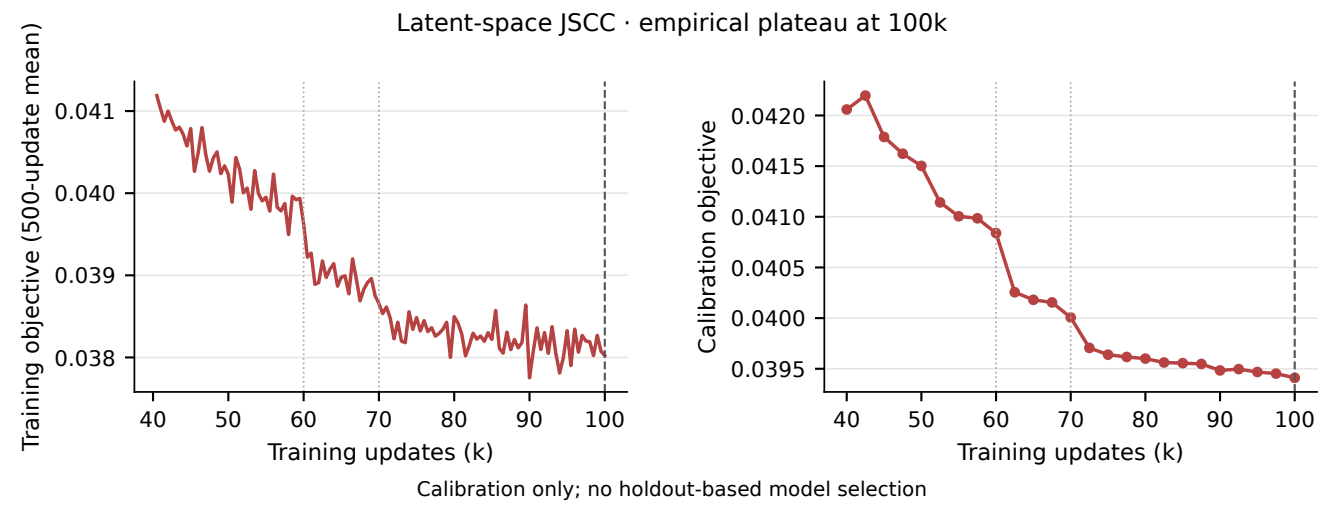

Partial-scale, no completion (raw)
NeST-Com (raw)
(a)
PSNR (dB) ↑
22
20
18
16
1
4
7
10
13
19
SNR (dB)
(b)
LPIPS ↓
0.6
0.4
0.2
SNR (dB)
(c)
DINOv2-L similarity ↑
0.8
0.6
0.4
0.2
0.0
SNR (dB)
(d)
Prediction agreement (%) ↑
75
50
25
0
SNR (dB)
Common 500 sources · N = 1024

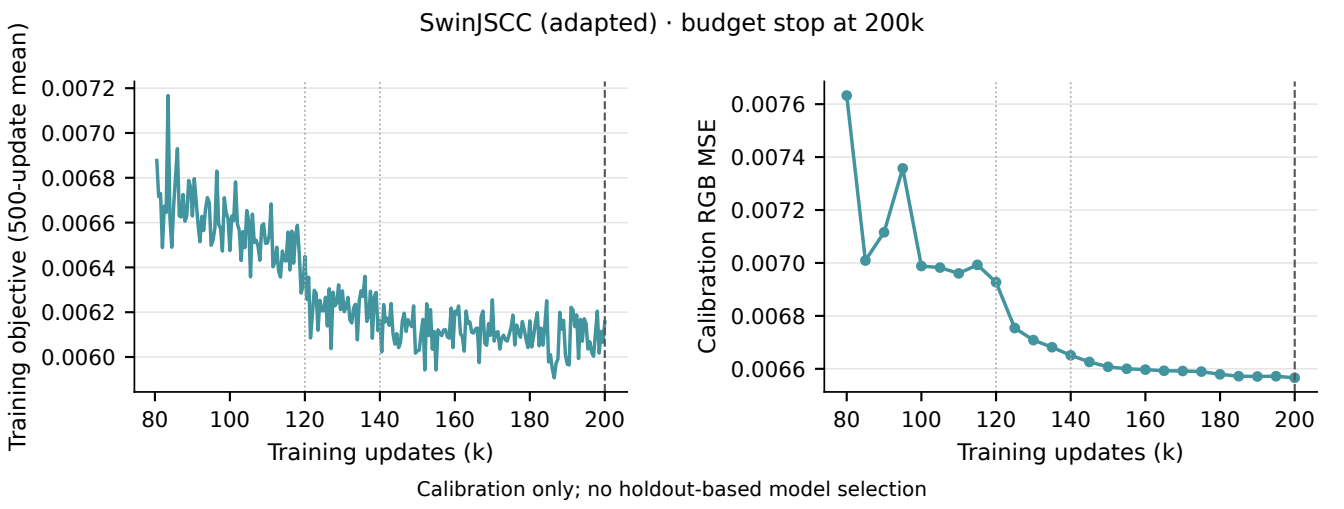

NeST-Com (entropy-coded)
SwinJSCC (adapted)
Latent-space JSCC
(a)
PSNR (dB) ↑
24
22
20
18
16
4
10
19
SNR (dB)
(b)
LPIPS ↓
0.4
0.2
4
10
19
SNR (dB)
(c)
DINOv2-L similarity ↑
0.8
0.6
0.4
0.2
4
10
19
SNR (dB)
(d)
Prediction agreement (%) ↑
100
80
60
40
20
4
10
19
SNR (dB)
Existing new100 sources · N = 1024

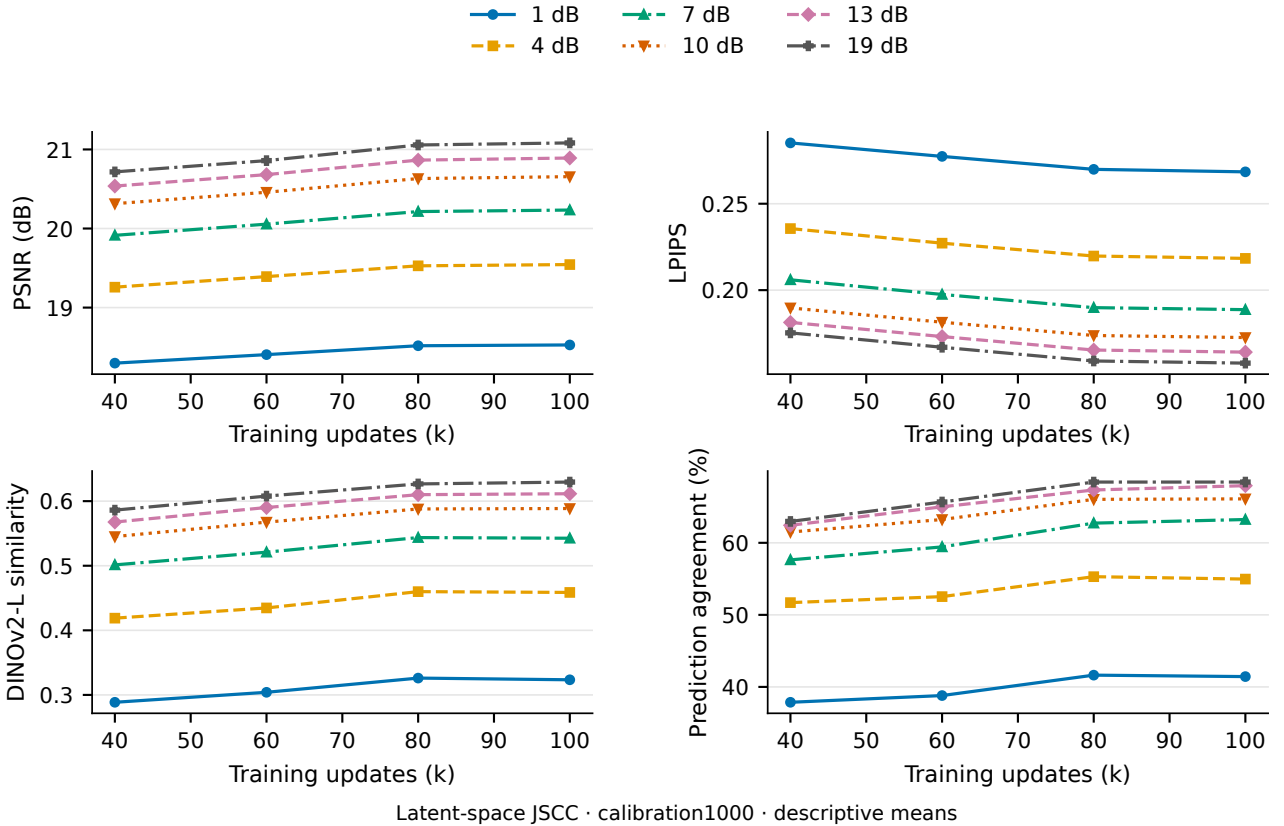

Whole-scale prefix + VAR (raw)
Whole-scale prefix + VAR (entropy-coded)
NeST-Com (raw)
(a)
PSNR (dB) ↑
24
22
20
18
4
10
19
SNR (dB)
(b)
LPIPS ↓
0.20
0.15
0.10
4
10
19
SNR (dB)
(c)
DINOv2-L similarity ↑
0.9
0.8
0.7
4
10
19
SNR (dB)
(d)
Prediction agreement (%) ↑
90
80
70
4
10
19
SNR (dB)
N2048 confirmation 100 · N = 2048

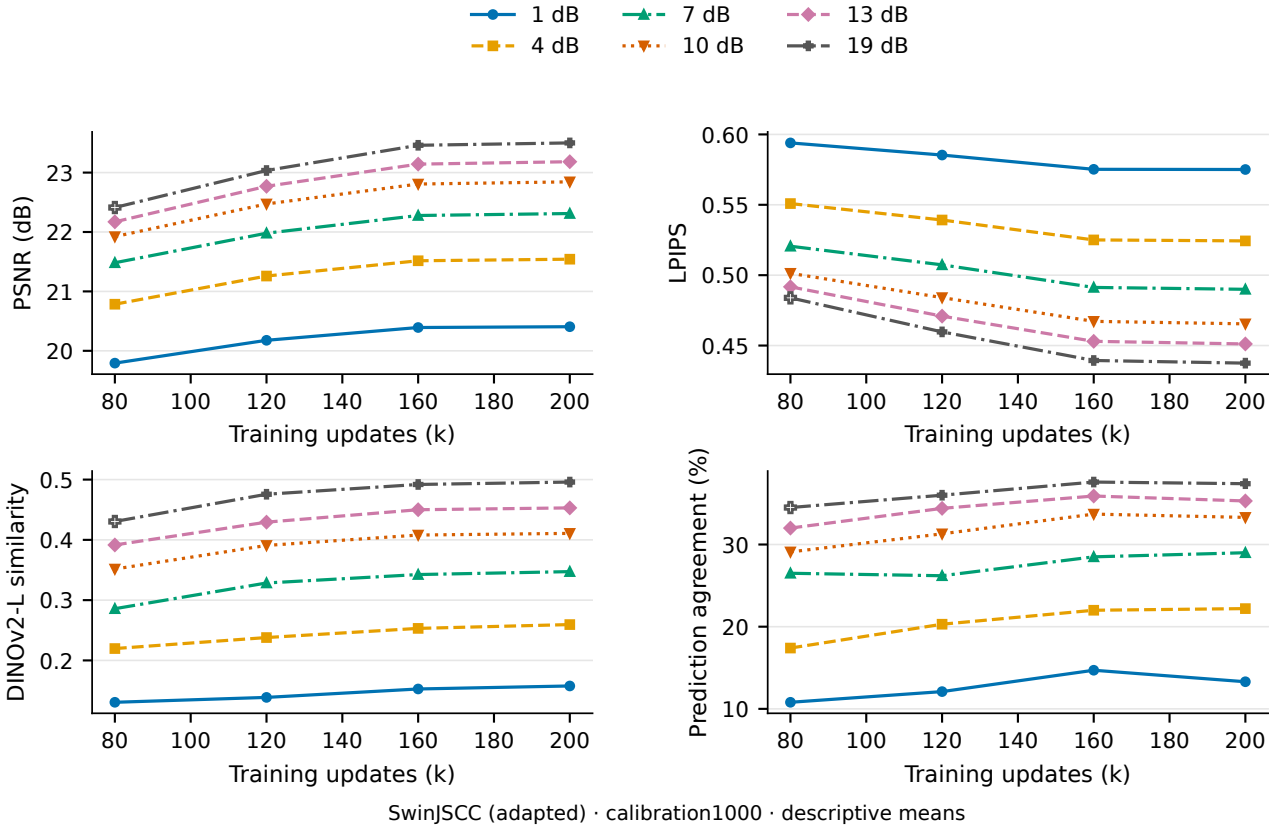

vs. Latent-space JSCC
vs. SwinJSCC (adapted)
NeST-Com (raw) − reference
(a)
ΔPSNR (dB) ↑
0
−2
−4
1
4
7
10
13
19
SNR (dB)
(b)
ΔLPIPS ↓
0.0
−0.1
−0.2
−0.3
SNR (dB)
(c)
ΔDINOv2-L similarity ↑
0.3
0.2
0.1
0.0
SNR (dB)
(d)
ΔPrediction agreement (pp) ↑
40
20
0
SNR (dB)
common500 · N = 1024

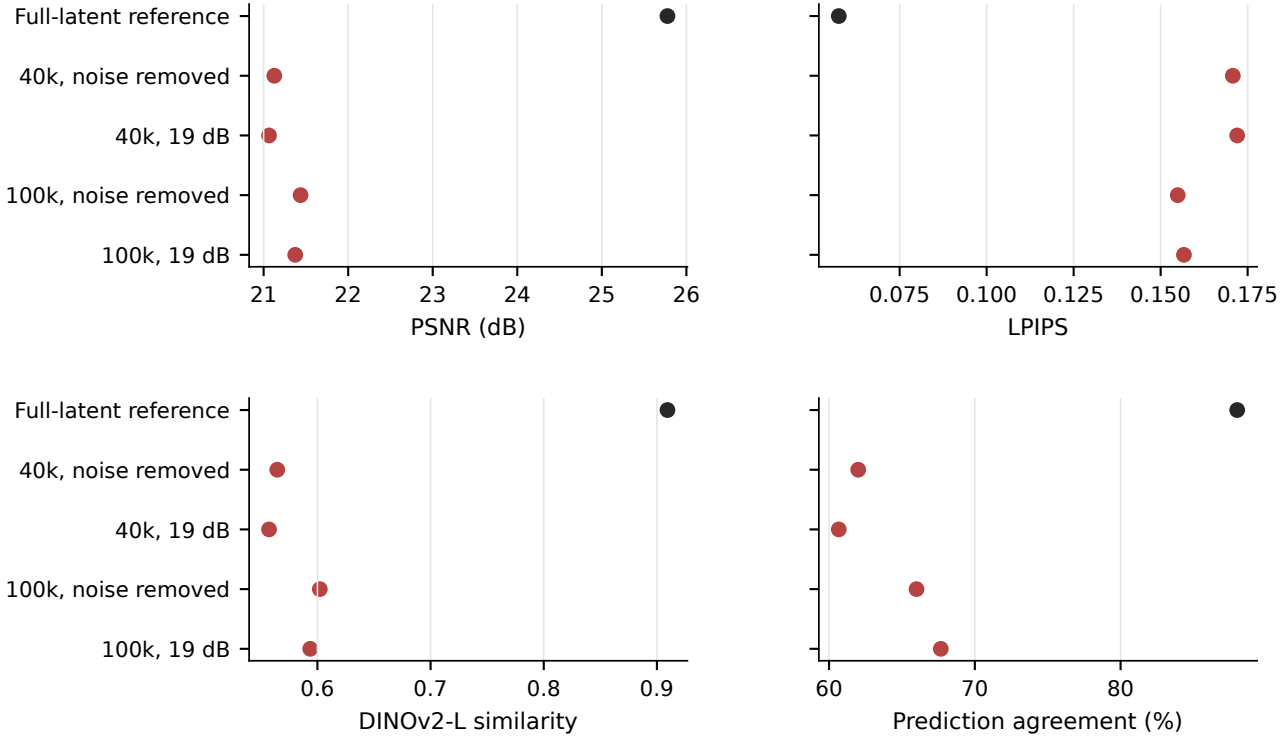

vs. Latent-space JSCC
vs. SwinJSCC (adapted)
NeST-Com (entropy-coded) − reference
(a)
ΔPSNR (dB) ↑
0
−2
−4
4
10
19
SNR (dB)
(b)
ΔLPIPS ↓
0.0
−0.1
−0.2
−0.3
4
10
19
SNR (dB)
(c)
ΔDINOv2-L similarity ↑
0.4
0.3
0.2
0.1
0.0
4
10
19
SNR (dB)
(d)
ΔPrediction agreement (pp) ↑
60
40
20
0
4
10
19
SNR (dB)
new100 · N = 1024


Latent-space JSCC · empirical plateau at 100k
Training objective (500-update mean)
0.041
0.040
0.039
0.038
40
50
60
70
80
90
100
Training updates (k)
Calibration objective
0.0420
0.0415
0.0410
0.0405
0.0400
0.0395
40
50
60
70
80
90
100
Training updates (k)
Calibration only; no holdout-based model selection

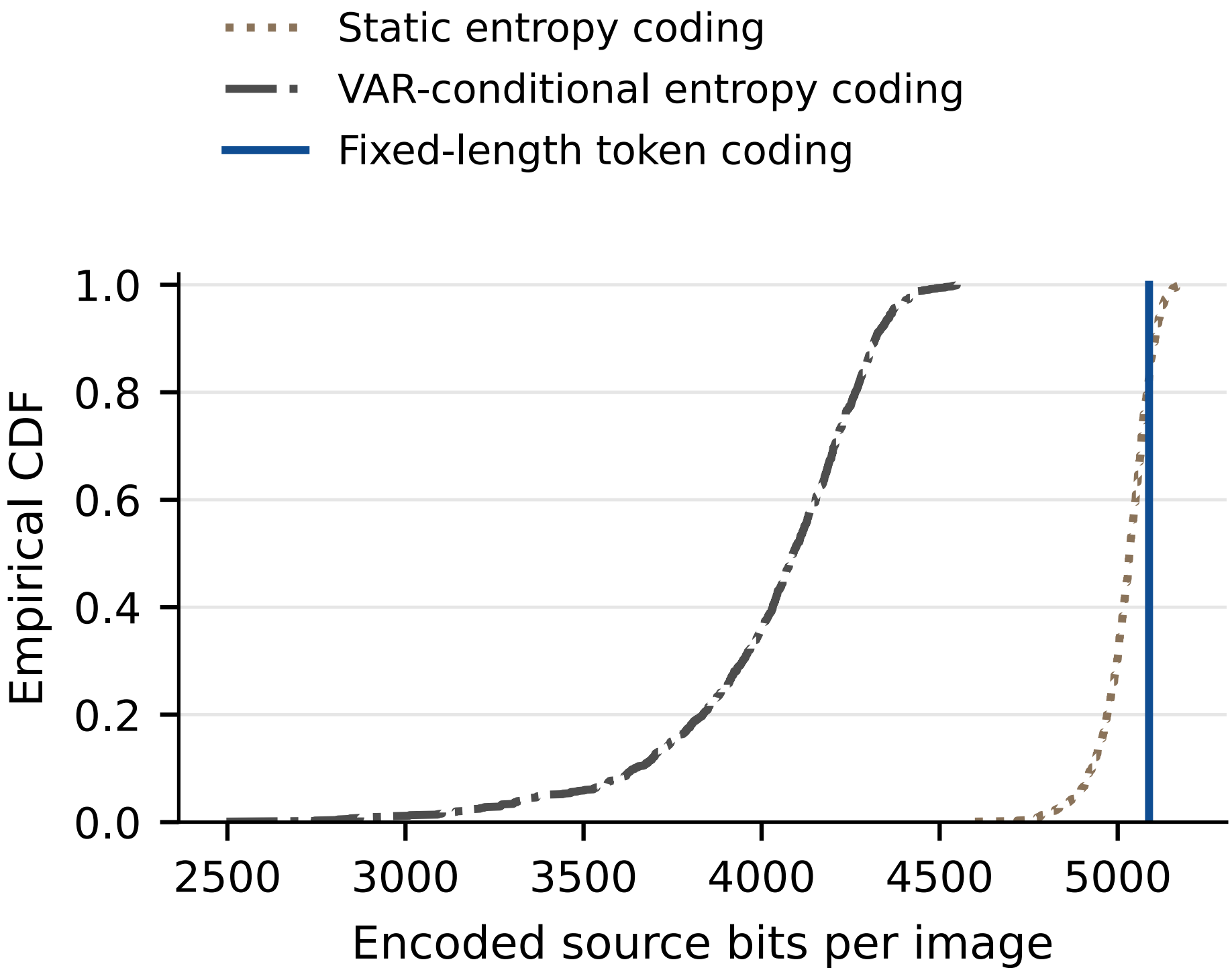

SwinJSCC (adapted) · budget stop at 200k
Training objective (500-update mean)
0.0072
0.0070
0.0068
0.0066
0.0064
0.0062
0.0060
80
100
120
140
160
180
200
Training updates (k)
Calibration RGB MSE
0.0076
0.0074
0.0072
0.0070
0.0068
0.0066
80
100
120
140
160
180
200
Training updates (k)
Calibration only; no holdout-based model selection


1 dB
4 dB
7 dB
10 dB
13 dB
19 dB
PSNR (dB)
21
20
19
LPIPS
0.25
0.20
DINOv2-L similarity
0.6
0.5
0.4
0.3
Prediction agreement (%)
60
50
40
40
50
60
70
80
90
100
Training updates (k)
Latent-space JSCC · calibration1000 · descriptive means

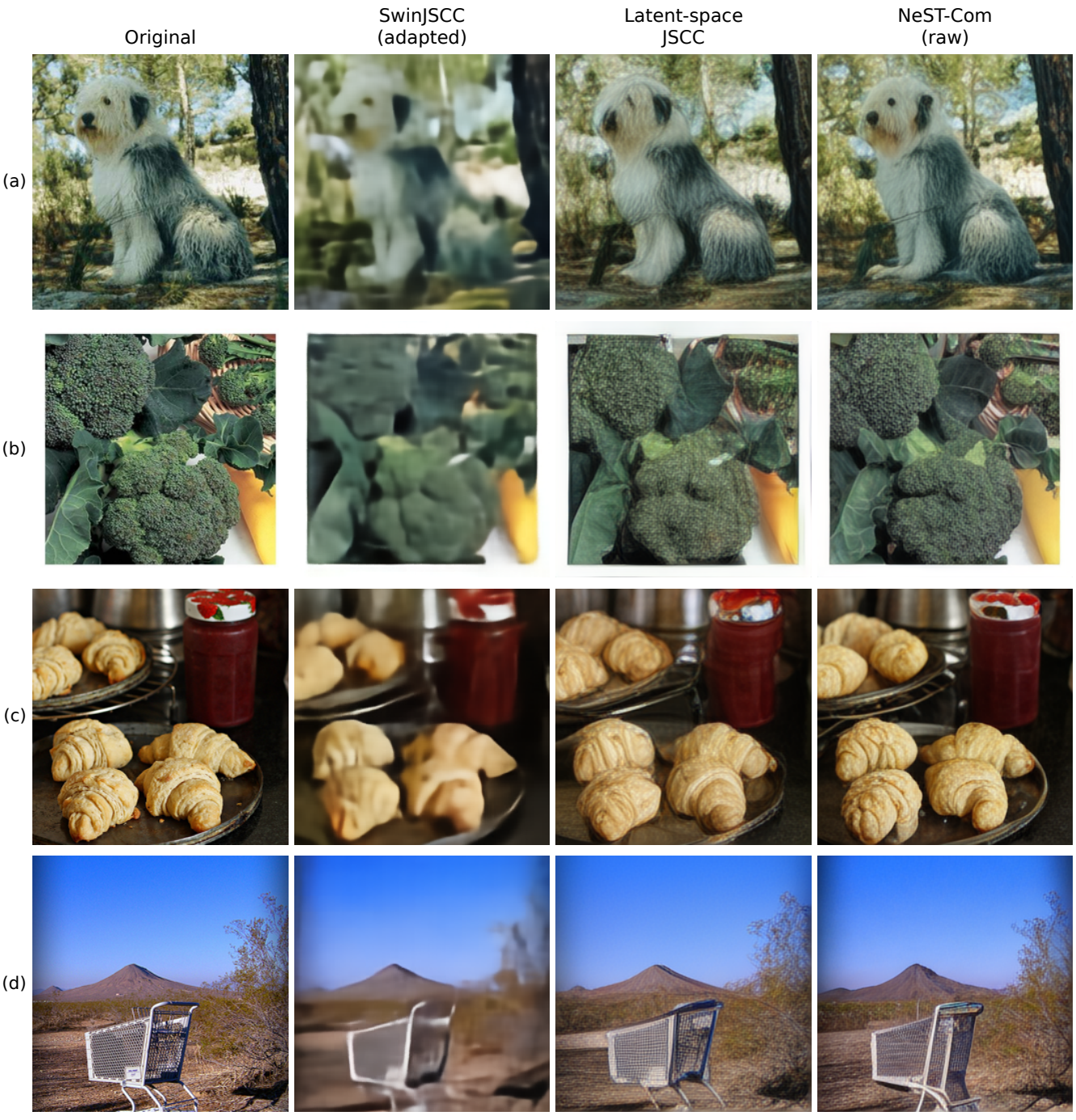

1 dB
4 dB
7 dB
10 dB
13 dB
19 dB
PSNR (dB)
LPIPS
DINOv2-L similarity
Prediction agreement (%)
Training updates (k)
SwinJSCC (adapted) · calibration1000 · descriptive means



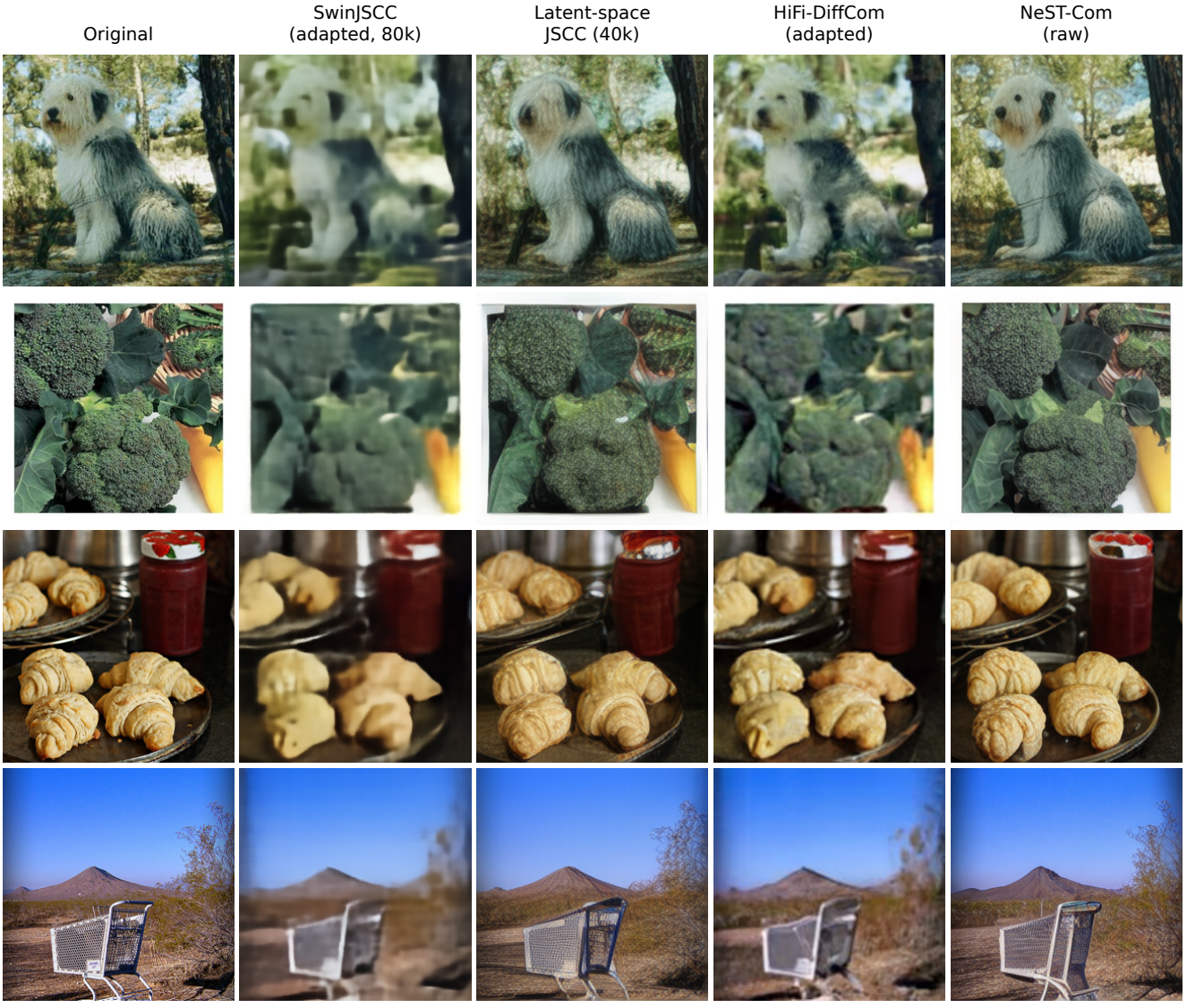
Latent-space JSCC bottleneck diagnostic · fixed calibration100 · descriptive means

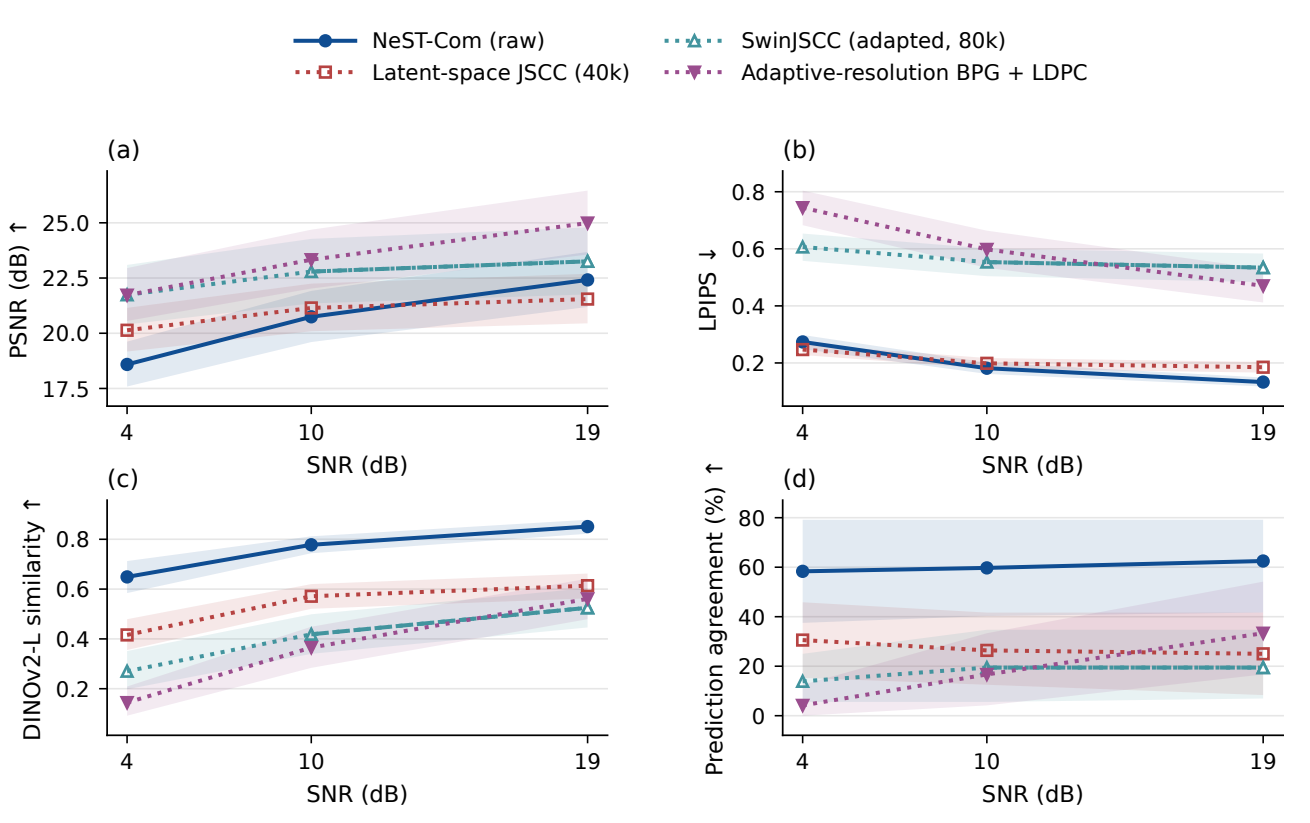

Static entropy coding
VAR-conditional entropy coding
Fixed-length token coding
Empirical CDF
1.0
0.8
0.6
0.4
0.2
0.0
2500
3000
3500
4000
4500
5000
Encoded source bits per image

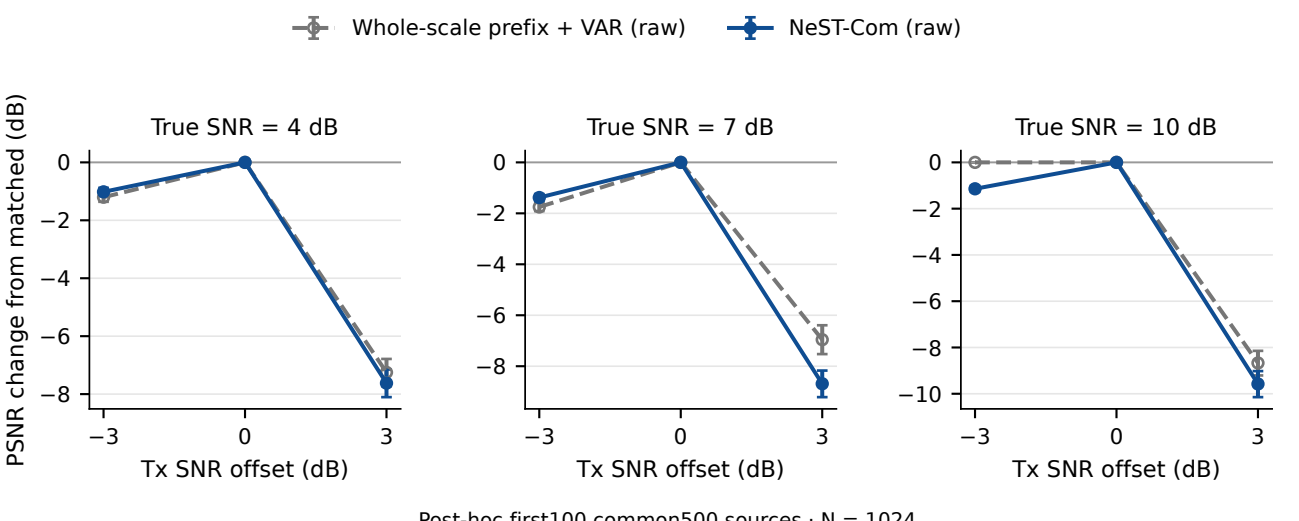

Original
SwinJSCC (adapted)
Latent-space JSCC
NeST-Com (raw)
(a)
(b)
(c)


Original
SwinJSCC (adapted)
Latent-space JSCC
NeST-Com (raw)
(a)
(b)
(c)
(d)


Original
Direct decoding
VAR completion


Original
SwinJSCC
(adapted, 80k)
Latent-space
JSCC (40k)
HiFi-DiffCom
(adapted)
NeST-Com
(raw)


NeST-Com (raw)
Latent-space JSCC (40k)
SwinJSCC (adapted, 80k)
Adaptive-resolution BPG + LDPC
(a)
PSNR (dB) ↑
25.0
22.5
20.0
17.5
4
10
19
SNR (dB)
(b)
LPIPS ↓
0.8
0.6
0.4
0.2
4
10
19
SNR (dB)
(c)
DINOv2-L similarity ↑
0.8
0.6
0.4
0.2
4
10
19
SNR (dB)
(d)
Prediction agreement (%) ↑
80
60
40
20
0
4
10
19
SNR (dB)
Kodak24 · 256×256 center crops · N = 1024

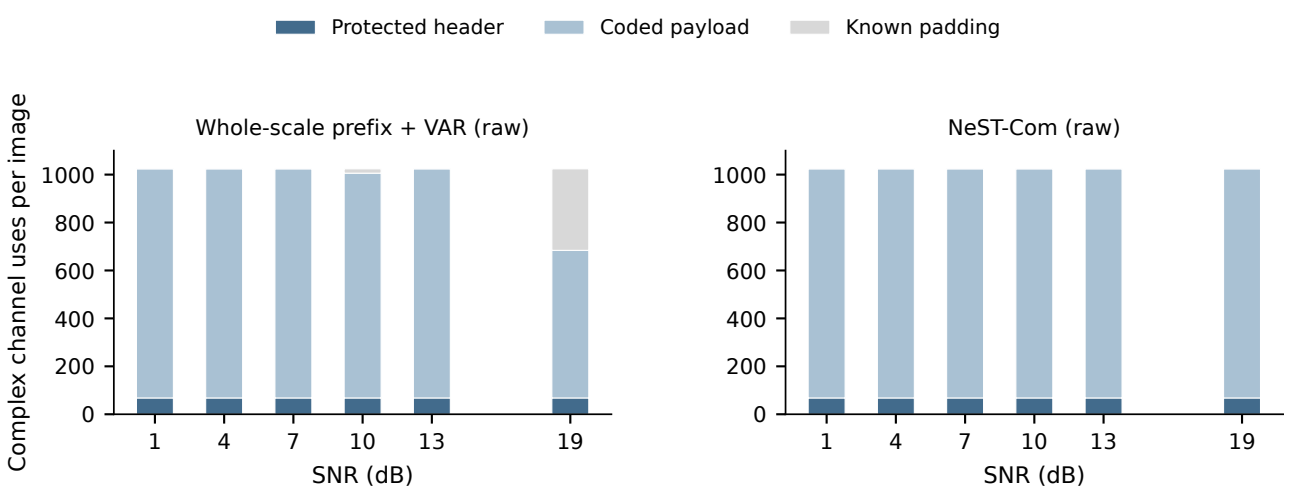

Whole-scale prefix + VAR (raw)
NeST-Com (raw)
True SNR = 4 dB
True SNR = 7 dB
True SNR = 10 dB
PSNR change from matched (dB)
Tx SNR offset (dB)
Tx SNR offset (dB)
Tx SNR offset (dB)
Post-hoc first100 common500 sources · N = 1024

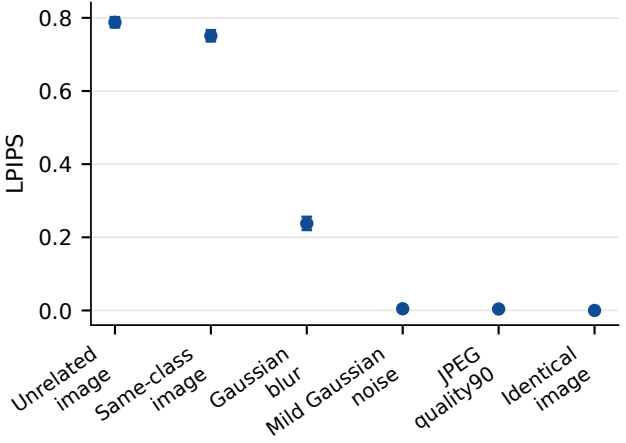
Whole-scale prefix + VAR (raw)

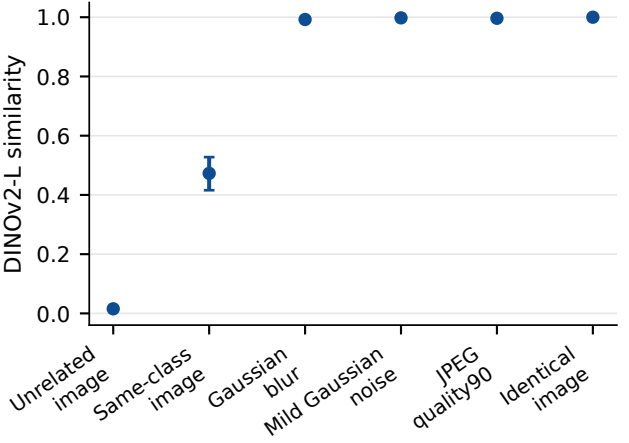
SNR (dB)	m	K	Modulation	k/n
1	5	0	QPSK	676/1912
4	6	0	16QAM	1108/3824
7	7	0	16QAM	1876/3824
10	7	0	64QAM	1876/5628
13	8	0	64QAM	3076/5736
19	8	0	64QAM	3076/3696

NeST-Com (raw)

SNR (dB)	m	K	Modulation	k/n
1	5	18	QPSK	892/1912
4	6	16	QPSK	1300/1912
7	7	0	16QAM	1876/3824
10	7	75	16QAM	2776/3824
13	8	9	16QAM	3184/3824
19	8	142	64QAM	4780/5736


Protected header
Coded payload
Known padding
Whole-scale prefix + VAR (raw)
NeST-Com (raw)
Complex channel uses per image
0
200
400
600
800
1000
1
4
7
10
13
19
SNR (dB)


LPIPS
0.0
0.2
0.4
0.6
0.8
Unrelated image
Same-class image
Gaussian blur
Mild Gaussian noise
JPEG quality90
Identical image


DINOv2-L similarity
0.0
0.2
0.4
0.6
0.8
1.0
Unrelated image
Same-class image
Gaussian blur
Mild Gaussian noise
JPEG quality90
Identical image